

Adjustment Sets and Approaches - and limitations / critiques

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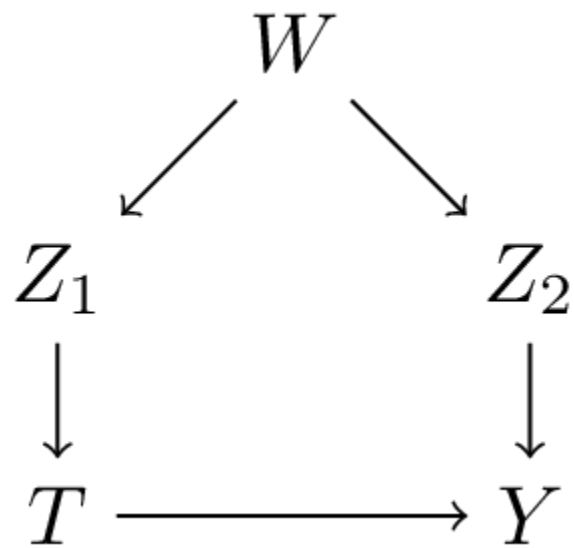


Adjustment sets and approaches

How to find adjustment sets?

- adjustment sets:
 - the back-door criterion states that any set Z that blocks all backdoor paths from X to Y is a sufficient adjustment set for causal effect estimation of $P(Y | \text{do}(X))$ using the backdoor formula.
 - how do we find these sufficient sets?
 - what if there are multiple?
- adjustment: how to do this?
 - stratification
 - what is regression adjustment?
 - T-learner vs S-learner

Valid adjustment sets



dag

- in general:
 - PA_T (the direct parents of treatment $T: Z_1$) are a valid adjustment set
 - PA_Y (the direct parents of outcome $Y: Z_2$) are a valid adjustment set
- in this case:
 - W is also a valid adjustment set

Valid adjustment sets: picking one

- websites like dagitty.net and causalfusion.net provide user-friendly interfaces for creating and exporting DAGs, in addition:
 - valid adjustment sets (if they exist)
 - testable conditional independencies

causalfusion.net
daggity.net



How to do adjustment



What not to do

1. do univariable pre-screening against outcome (and / or treatment)
 - this should maybe never be done
 - especially not in the context of causal inference

Adjustment formula

$$P(y|\text{do}(x)) = \sum_z P(y|x, z)P(z)$$

- entails summing over all possible values of Z
- say Z is 5 categorical variables with each 3 categories, this means $3^5 = 243$ estimates of:
 - $P(y|x, z)$ for each value of x
- what if Z is continuous?
- in practice, researchers often rely on smoothness assumptions (e.g. regression) to estimate $P(Y|x, z)$ with a parametric model
- this assumption *can* be based on substantive causal knowledge, but often seems inspired rather pragmatism or necessity
- misspecification of this estimator leads to biased results (even if you observed all the confounders)

Target queries

- up to now we've worked exclusively with $P(y|\text{do}(t))$: the probability of observing outcome y when setting treatment T to t
- this is not typically what is of most interest, say there are two treatment options $T \in \{0, 1\}$ (control and 'treatment')

1. *average treatment effect*

$$\text{ATE} = E[y|\text{do}(t = 1)] - E[y|\text{do}(t = 0)]$$

2. *conditional average treatment effect*

$$\text{CATE} = E[y|\text{do}(t = 1), w] - E[y|\text{do}(t = 0), w]$$

3. *prediction-under-intervention* $P(y|\text{do}(t), w)$ (more on this on [day 4](#))

- these can be computed from $P(y|\text{do}(t), w)$

The simplest case: linear regression

- assume the following structural causal model (z is confounder, u is exogenous noise):
- then:
- i.e. the ATE collapses to the regression parameter in a linear regression model of y on x

Binary outcome: logistic regression

- researchers make the following substantive assumption: the log-odds of the outcome is linear in the treatment and confounders:
- then the ATE is not equal to anymore, but can be computed from the regression model:
- after fitting a logistic regression model in R with `glm`, the ATE can be computed with the package `marginalEffects` (or `margins`)

General estimators for the ATE and the CATE (meta-learners)

- denote
 - (assuming is a sufficient set)
- T-learner: model and separately (e.g. regression separately for treated and untreated):
- S-learner: use as just another feature
- (many other variants combinations: this is a whole literature)

Doubly robust estimators

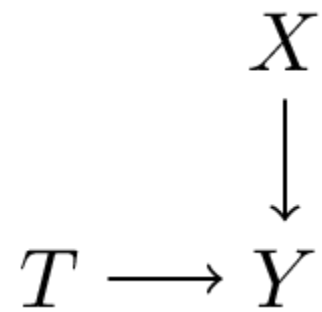
- In day 1, we saw that the ATE can be estimated with the propensity score:
- Doubly robust estimators combine the propensity score and outcome regression:
- They have the property that if either the propensity score model or the outcome regression model is correctly specified, the estimator is consistent for the ATE.
- In addition, they may be more *efficient* than either method alone, especially when both models are correctly specified.

How to choose between estimators?

Intuitive way-pointers:

- where does the complexity come from?
 - a. variance in outcome under control:
 - b. variance CATE: (in statistics: *interaction* between treatment and covariate)

Where does the variance come from?



DAG

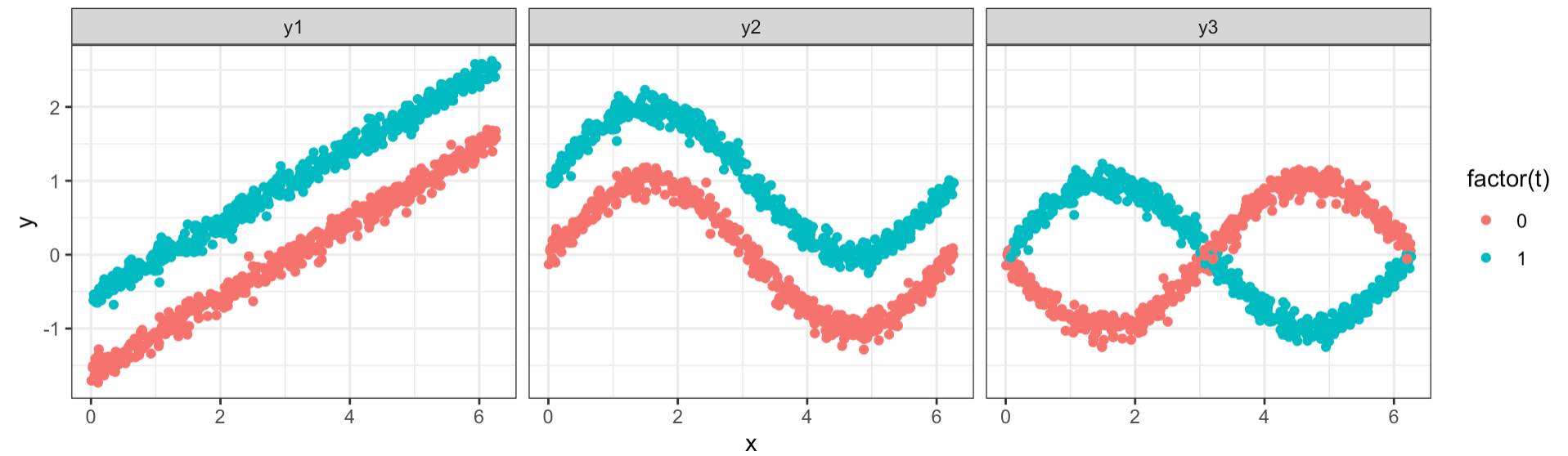


Figure 1: Three datasets with the same DAG

1. (linear)
2. (non-linear additive)
3. (non-linear + interaction)

Limitations of DAGs and SCMs

Making DAGs

- how do you get a DAG? up to now we assumed we had one
- based on prior evidence, expert knowledge
- “no causes in, no causes out”

A003024: The death of DAGs?

The number of possible DAGs grows super-exponentially in the number of nodes

n_nodes	n_dags	time at 1 sec / DAG
1	1	
2	3	
3	25	
4	543	
5	29281	> an hour
6	3781503	> a day
7	1138779265	> a year
8	783702329343	
9	1213442454842881	> human species
10	4175098976430598143	> age of universe



Do we need to consider all DAGs?

- a single sufficient set suffices
- adjusting for all direct causes of the treatment or all direct causes of the outcome are always sufficient sets
- can we judge these without specifying all covariate-covariate relationships?
- potential approach:
 - put all potential confounders in a cluster (e.g [Anand et al. 2023](#))
 - ignore covariate-covariate relationships in that cluster
 - what happens when (partial) missing data?

SCM vs potential outcomes

- definition of causal effect
 - PO: averages of individual potential outcomes
 - SCM: submodel or mutilated DAG
- both require positivity
- d-separation implies conditional independence (exchangeability)

References

Anand, Tara V., Adele H. Ribeiro, Jin Tian, and Elias Bareinboim. 2023. “Causal Effect Identification in Cluster DAGs.” *Proceedings of the AAAI Conference on Artificial Intelligence* 37 (10): 12172–79. <https://doi.org/10.1609/aaai.v37i10.26435>.